PHYSICS

1) A small block of mass 2 kg is kept on a rough inclined surface of inclination of 30° fixed in a lift. The lift goes up with a uniform speed of 1 ms^{-1} and the block does not slide relative to the inclined surface. The work done by the force of friction on the block in a time interval of 2s is :-

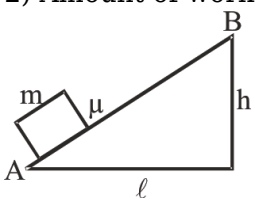
(1) zero

(2) 9.8 J

(3) 29.4 J

(4) 16.9 J

2) Amount of work done to carry a block from A to B will be (Assume friction coefficient μ)



(1) mgh

(2) $\frac{\mu mg}{\sqrt{\ell^2 + h^2}}$

(3) μmg(+ h)

(4) mg(h + μ)

3) Force acting on a particle moving in a straight line varies with the velocity of the particle as $F = \frac{k}{v}$, where k is a constant. The work done by this force in time t is :-

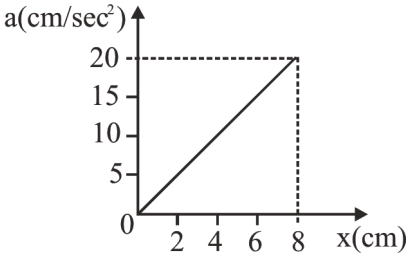
(1) $\frac{k}{v^2}t$

(2) 2kt

(3) kt

(4) $\frac{2kt}{v^2}$

4) A 5 kg mass moves along x-axis. Its acceleration as a function of its position is as shown in the figure. What is the total work done on the mass by the force, as the mass moves from x = 0 cm to x = 8 cm ?



(1) 4×10^{-2} J

(2) 8×10^{-2} J

(3) 2×10^{-2} J

(4) 0.8×10^{-2} J

5) A time dependent force F = 6t acts on a particle of mass 1kg. If the particle starts from rest. The work done by the force during the first 1sec. will be :

(1) 9 J

(2) 4.5 J

(3) 18 J

(4) 22 J

6) Consider the following two statements :- A : Linear momentum of a system of particles is zero.
B : Kinetic energy of a system of particles is zero.
Then-

(1) A does not imply B and B does not imply A

(2) A implies B but B does not imply A

(3) A does not imply B but B implies A

(4) A implies B and B implies A

7) Because of a force, position of a particle is given by $x = (t - 3)^2$; then find work done in first six seconds. (mass of particle = m)

(1) 18m

(2) 0

(3) $\frac{9m}{2}$

(4) 36m

8) A small block released with a horizontal velocity v_0 on a rough horizontal surface, comes to rest after covering a distance d, the speed of the block when it had covered a distance d/4 is :-

(1) $\frac{v_0}{2}$

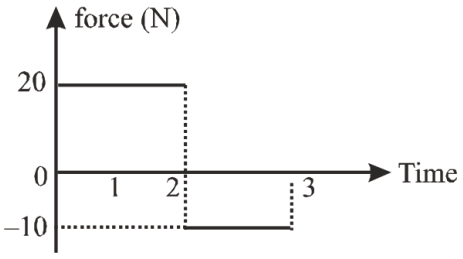
(2) $\frac{v_0}{4}$

(3) $\frac{\sqrt{3}v_0}{2}$

(4) $\frac{\sqrt{3}v_0}{4}$

9) Starting at rest, a 5 kg object is acted upon by only one force as indicated in graph. Find the total

work done by the force.



(1) 90 J
(2) 80 J
(3) 120 J
(4) 100 J

10) Which of the following statement is false :

(1) Work done by conservative force is path independent
(2) If only conservative forces are present then mechanical energy is conserved
(3) All conservative forces follows inverse square law
(4) Work done by conservative force in a closed path is zero

11) The potential energy function associated with the force $\vec{F} = 4xy\hat{i} + 2x^2\hat{j}$ is :-

(1) $U = -2x^2y$
(2) $U = -2x^2y + c$
(3) $U = 2x^2y + c$
(4) None of these

12) A chain of mass m and length is held on a frictionless table in such a way that its $\frac{2\ell}{5}$ part is hanging below the edge of table. Find work done to pull the hanging part of chain.

(1) $\frac{2mg\ell}{25}$
(2) $\frac{mg\ell}{50}$
(3) $\frac{mg\ell}{15}$
(4) $\frac{4mg\ell}{15}$

13) A particle is released from a height S. At certain height its kinetic energy is three times its potential energy. The height and speed of the particle at that instant are respectively :-

(1) $\frac{S}{4}, \frac{3gS}{2}$
(2) $\frac{S}{4}, \frac{\sqrt{3gS}}{2}$

(3) $\frac{S}{2}, \frac{\sqrt{3gS}}{2}$

(4) $\frac{S}{4}, \sqrt{\frac{3gS}{2}}$

14) A 15 g ball is shot from a spring gun whose spring has a force constant of 600 N/m. The spring is compressed by 3 cm. The greatest possible velocity of the ball for this compression is($g = 10\ m/s^2$)–

(1) 6.0 m/s

(2) 12.0 m/s

(3) 10.0 m/s

(4) 8.0 m/s

15) An electric pump is used to fill an overhead tank of capacity $9m^3$ kept at a height 10m above ground. If the pump takes 5 minutes to fill the tank by consuming 10 kW power, efficiency of the pump should be : ($g = 10\ m/s^2$) :-

(1) 60%

(2) 40%

(3) 20%

(4) 30%

16) If position of a particle of mass 1kg varies with time is given by $x = t^2 + 2t - 1$, then average power developed in 1 second :-

(1) 6W

(2) 8W

(3) 10W

(4) 12W

17) Select the correct statement from the following :

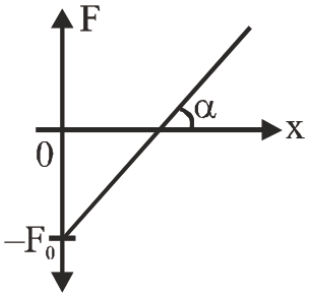
(1) Work done by static friction is always zero

(2) Work done by normal contact force is always zero

(3) Work done by kinetic friction may be positive

(4) Net work done by internal conservative force depends on frame of reference.

18) An object starts from rest and is acted upon by a variable force F as shown in figure. If F_0 is the initial value of the force, then the position of the object, where it again comes to rest will be :-



(1) $\frac{2F_0}{\tan\alpha}$

(2) $\frac{F_0}{\sin\alpha}$

(3) $\frac{2F_0}{\cot\alpha}$

(4) $\frac{F_0}{2\cos\alpha}$

19) The ratio of momentum and kinetic energy of particle is inversely proportional to time. Then this is the case of a :

(1) uniformly accelerated motion

(2) uniformly motion

(3) uniformly retarded motion

(4) simple harmonic motion

20) A particle of mass 1kg is projected vertically upwards with a speed of 16 ms^{-1}. After some time, when it again passes through the point of projection, its speed is found to be 10 ms^{-1}. then work done by air resistance during this motion is:

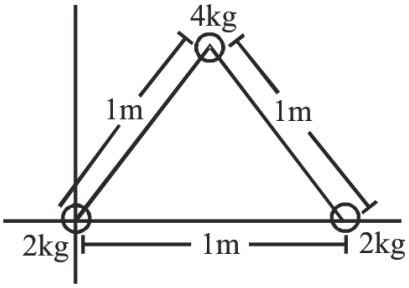
(1) –128 J

(2) –256 J

(3) –78 J

(4) –50 J

21) Co-ordinates of centre of mass of given system are :-



(1) $\left(\frac{1}{2}, \frac{\sqrt{3}}{4}\right)$

(2) $\left(\frac{1}{2}, 2\sqrt{3}\right)$

(3) $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$

(4) $\left(\frac{1}{2}, 3\sqrt{3}\right)$

22) Blocks A and B are resting on a smooth horizontal surface given equal speeds of 2 m/s in



opposite sense as shown in the figure. At t = 0 the position of blocks are shown, then the co-ordinates of center of mass at t = 3s will be :-

(1) (1, 0)

(2) (3, 0)

(3) (5, 0)

(4) (2.25, 0)

23) **Assertion (A) :-** In case of bullet fired from gun, the ratio of kinetic energy of gun and bullet is equal to ratio of mass of bullet and gun.
Reason (R) :- In firing, total momentum is conserved.

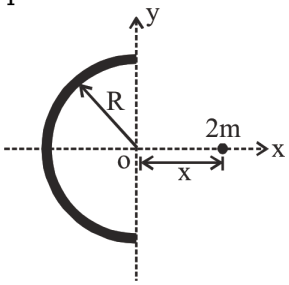
(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**

(2) **(A)** is correct but **(R)** is not correct

(3) **(A)** is incorrect but **(R)** is correct

(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**

24) A ring of mass m and a particle of mass 2m are placed as shown in figure. If the centre of mass



of the combination lies at origin, the value of x is :-

(1) $\frac{2R}{\pi}$

(2) $\frac{R}{\pi}$

(3) $\frac{R}{2\pi}$

(4) $\frac{2R}{3\pi}$

25) The centre of mass of a non uniform rod of length L whose mass per unit length varies as $\rho = \frac{kx^3}{L^2}$ (where k is a constant and x is the distance measured from one end) is at the following distance from the same end.

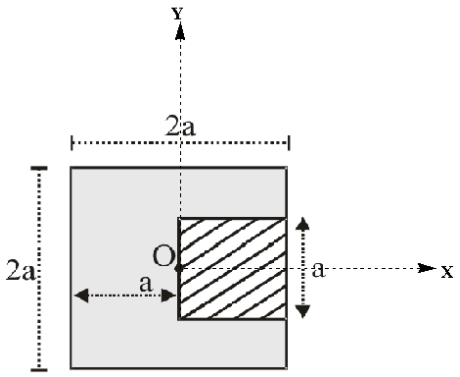
(1) 3L/4

(2) 4L/5

(3) 2L/3

(4) L/3

26) A plate of uniform thickness is shown in figure. It's center of mass is at origin O. Now if the shaded area is removed, the new position of the centre of mass is :-



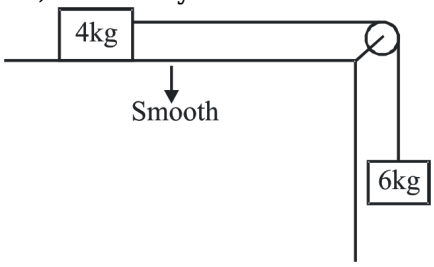
(1) $\frac{a}{6}\hat{i}$

(2) $\frac{a}{3}\hat{i}$

(3) $\frac{-a}{3}\hat{i}$

(4) $\frac{-a}{6}\hat{i}$

27) For the system shown in the figure the acceleration of centre of mass is :-



(1) $\left(2.4\hat{i}-3.6\hat{j}\right)\ m/s^2$

(2) $\left(2.4\hat{i}+3.6\hat{j}\right)\ m/s^2$

(3) $\left(3.6\hat{i}-2.4\hat{j}\right)\ m/s^2$

(4) $\left(3.6\hat{i}+2.4\hat{j}\right)\ m/s^2$

28) Two particles are shown in figure. At t = 0, a constant force F = 6N starts acting on the 3 kg mass. Find the velocity of the centre of mass of these particles at t = 5s.



(1) 5 m/s

(2) 4 m/s

(3) 6 m/s

(4) 3 m/s

29) A stationary body explodes into-four identical fragments such that three of them fly off mutually perpendicular to each other, each of three with same kinetic energy E_0, Then the energy of explosion will be :

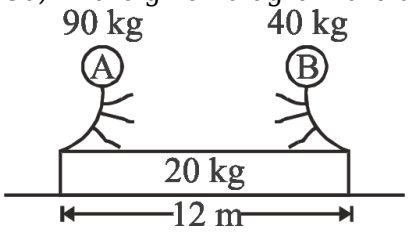
(1) $6E_0$

(2) $\frac{4E_0}{3}$

(3) $4E_0$

(4) $8E_0$

30) In the given diagram the distance travelled by the plank when A and B exchange their positions:



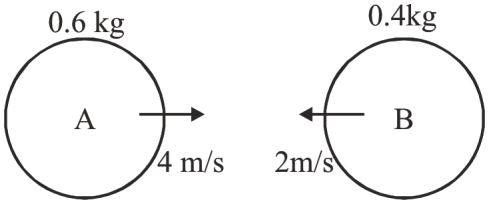
(1) 2 m

(2) 4 m

(3) 4.8 m

(4) 5.4 m

31) The velocities of two steel balls before impact are shown. If after head on impact the velocity of ball B is observed to be 3 m/s to the right, the coefficient of restitution is



(1) $\frac{7}{18}$

(2) $\frac{6}{7}$

(3) $\frac{5}{18}$

(4) None of these

32) Which of the following statement is incorrect:

(1) Law of conservation of linear momentum is valid in all inertial reference frame

(2) Linear momentum is independent of reference frame

(3) Total momentum of system is zero in centre of mass frame

(4) Kinetic energy depends on frame of reference

33) Ball 1 collides head on with an another identical ball 2 at rest. Velocity of ball 2 after collision becomes two times to that of ball 1 after collision. The coefficient of restitution between the two ball is :-

(1) $e = \frac{1}{3}$

(2) $e = \frac{1}{2}$

(3) $e = \frac{1}{4}$

(4) $e = \frac{2}{3}$

34) A ball hits the ground with speed 10 m/s at an angle of 45° with the ground then rebound velocity will be, if coefficient of restitution $e = \frac{1}{\sqrt{2}}$:-

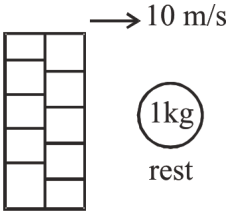
(1) $10\sqrt{2}$ m/s

(2) 10 m/s

(3) $5\sqrt{3}$ m/s

(4) 5 m/s

35) Consider the situation as shown in the diagram. The wall strikes the stationary ball normally and collision is elastic. The speed of ball after collision is :



(1) 10 m/s

(2) 15 m/s

(3) 20 m/s

(4) None of these

36) A 4kg ball moving with speed 6 m/s collides to 1 kg ball moving with 1 m/s in same direction. If $e = \frac{1}{2}$ then the loss in kinetic energy during collision will be:

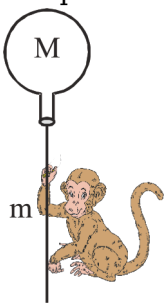
(1) 5.4 J

(2) 7.5 J

(3) 12.5 J

(4) None of these

37) A balloon of mass M with a light rope and monkey of mass m are at rest in mid air. If the monkey climbs up the rope and reaches the top of the rope, the distance by which the balloon descends will be (The length of the rope = L) :-



(1) $\frac{m}{(m+M)^2}$

(2) $\frac{mL}{m+M}$

(3) $\frac{(m+M)L}{m}$

(4) $\frac{(M+m)}{mL}$

38) Which of the following statements is correct ?

(1) Kinetic energy of a system can be changed without changing its momentum.

(2) Kinetic energy of a system cannot be changed without changing its momentum.

(3) Momentum of a system cannot be changed without changing its kinetic energy.

(4) Body cannot have energy without having momentum.

39) A body is dropped from a height of 8 m. After striking the surface it rises to 6 m what is fractional loss in kinetic energy during impact? (Assuming air resistance to be negligible) :-

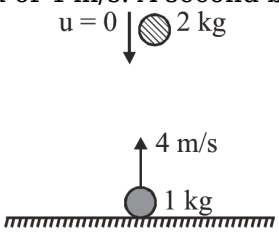
(1) 2/5

(2) 1/4

(3) 3/4

(4) 1/5

40) A ball of mass 1 kg is thrown up with an initial speed of 4 m/s. A second ball of mass 2 kg is released from rest from some height as shown in figure.



Consider the following statements :
(a) The centre of mass of the two balls comes down with acceleration g/3.
(b) The centre of mass first moves up and then comes down.
(c) The acceleration of the centre of mass is g downwards.
(d) The centre of mass of the two balls remains stationary.
Correct statements are :-

(1) only d

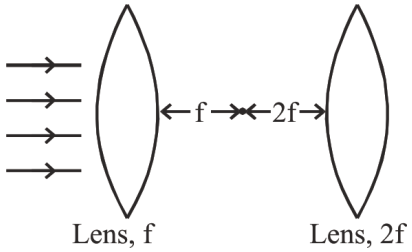
(2) only a

(3) a, b

(4) b, c

41) Two convex lens of focal length f and 2f are arranged to have their foci at the same point as shown in figure. A circular beam of parallel light rays having 1 cm cross section radius is incident on

first lens. The beam exiting the second lens is:-



(1) Diverging

(2) Converging

(3) Parallel and 1 cm cross section radius

(4) Parallel and of 2 cm cross section radius

42) The focal lengths of the objective and eye lenses of a telescope are respectively 200 cm and 5 cm. The maximum magnifying power of the telescope will be :-

(1) – 40

(2) – 48

(3) – 60

(4) – 100

43) A ray of light is incident on a surface of glass slab at an angle 45°. If the lateral shift produced per unit thickness is $\frac{1}{\sqrt{3}}$ m, the angle of refraction produced is :-

(1) $\tan^{-1}\left(\frac{\sqrt{3}}{2}\right)$

(2) $\tan^{-1}\left(1-\sqrt{\frac{2}{3}}\right)$

(3) $\sin^{-1}\left(1-\sqrt{\frac{2}{3}}\right)$

(4) $\tan^{-1}\left(\sqrt{\frac{2}{\sqrt{3}-1}}\right)$

44) A point object is placed on the principal axis of a concave mirror quite far away from the pole and moved at a constant speed 0.5 cm/sec towards the pole. The image also moves. It is found that the object and the image cross each other at a point which is at a distance 50 cm from the pole. Focal length of the mirror is -

(1) 50 cm

(2) 35 cm

(3) 25 cm

(4) 15 cm

45) Read the following statements :
(a) Both concave and convex mirror can form **Virtual** image for Real object.

(b) A **Virtual** image cannot be caught on a screen but it can be photographed.
(c) The **diver** under water looks obliquely, at a fisherman standing on the bank of a lake, the fisherman look taller to the diver, than what he actually is
(d) The refractive index of diamond is much greater than that of **original** glass
select the correct statements :

(1) a,b
(2) b,c
(3) a,b,c,d
(4) a,b,d

46) A Square wire frame of side 3 cm is placed 25 cm away from a concave mirror of focal length 10 cm. What is the area enclosed by the image of the wire? (The centre of the square wire is on the axis of the mirror and plane is perpendicular to principal axis) :-

(1) 2 cm^2
(2) 4 cm^2
(3) 9 cm^2
(4) None of these

47) **Assertion :** Power of plane mirror in air is P if plane mirror is placed in liquid of R.I. μ then power will not change.
Reason : Power of plane mirror is zero and it does not depend on surrounded medium.

(1) Both A and R are true and R is correct explanation of Assertion
(2) Both A and R are true and R is not correct explanation of Assertion
(3) Assertion is true and Reason is false
(4) Both Assertion and Reason are false

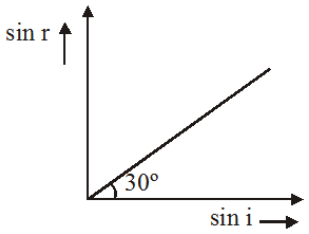
48) If power of spherical mirror is 5D then nature and radius of curvature of mirror is :

(1) Convex, –40 cm
(2) Concave, +20 cm
(3) Convex, –20 cm
(4) Concave, –40 cm

49) Monochromatic light of wavelength 589 nm is incident from air on a water surface. The refractive index of water is 1.33. The wavelength of the refracted light is :-

(1) 589 nm
(2) 443 nm
(3) 333 nm
(4) 221 nm

50) A medium shows the relation between the angle of incidence i and the angle of refraction r. If



speed of light in the medium is nc then value of n is

(1) 1.5

(2) 2

(3) 2^{-1}

(4) $3^{-1/2}$

51) When a ray is refracted from one medium to another, the wavelength changes from 6000 Å to 4000 Å. The critical angle for the interface will be

(1) $\cos^{-1}\left(\frac{2}{3}\right)$

(2) $\sin^{-1}\left(\frac{2}{\sqrt{3}}\right)$

(3) $\sin^{-1}\left(\frac{2}{3}\right)$

(4) $\cos^{-1}\left(\frac{2}{\sqrt{3}}\right)$

52) A light ray is incident on a rectangular slab of refractive index $\sqrt{2}$ at an angle of incidence 45° The angle between the reflected and refracted ray is

(1) 75°

(2) 90°

(3) 105°

(4) 120°

53) The velocity of light in glass is 2×10^8 m/sec and that in air is 3×10^8 m/sec. By how much would an ink dot appear to be raised, when covered by a glass plate 6 cm thick.

(1) 2 cm

(2) 4 cm

(3) 3 cm

(4) 6 cm

54) An air bubble in a glass slab with refractive index 1.5 (near normal incidence) is 5 cm deep when viewed from one surface and 3 cm deep when viewed from the opposite face. The thickness (in cm) of the slab is :-

(1) 12

(2) 16

(3) 8

(4) 10

55) How much water would be filled in a container of height 28 cm, so that it appears half filled to the observer when viewed from the top of the container ($\mu_w = 4/3$)

(1) 14 cm

(2) 16 cm

(3) 12 cm

(4) 10 cm

56) A fish is at the depth of 30 cm below the surface of water. If refractive index of water is 4/3 then find distance between the fish and its image :-

(1) 7.5 cm

(2) 5 cm

(3) 2.5 cm

(4) 10 cm

57) Critical angle of light passing from glass to air is minimum for :-

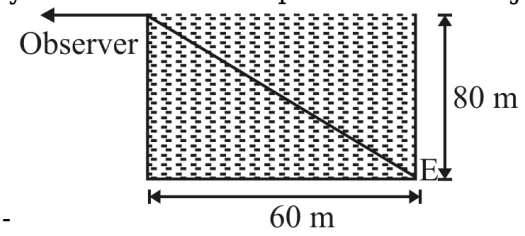
(1) red

(2) green

(3) yellow

(4) violet

58) When the rectangular metal tank is filled up to the top with an unknown liquid, an observer with eyes level with the top of the tank can just see the corner E. Find the index of refraction of the liquid



:-

(1) 1.5

(2) 1.25

(3) 1.67

(4) 2.4

59) A light ray is incident at an angle 30º on a transparent surface separating two media. If the angle of refraction is 60º then critical angle is

(1) $\sin^{-1}\left(\frac{2}{3}\right)$

(2) $\sin^{-1}\left(\sqrt{3}\right)$

(3) $\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$

(4) 45°

60) A, B and C are three optical media of respective critical angles C_1, C_2 and C_3. Total internal reflection of light can occur from A and B and also from B to C but not from C to A. Then the correct relation between critical angles is

(1) $C_1 > C_2 > C_3$

(2) $C_1 = C_2 = C_3$

(3) $C_3 > C_1 > C_2$

(4) $C_1 < C_2 < C_3$

61) A fish looking up through water sees the outside world contained in a circular horizon. If refractive index of water is $\frac{4}{3}$ and fish is 12 cm below the surface, radius of circle is :-

(1) $12 \times 3 \times \sqrt{5}$cm

(2) $12 \times 3 \times \sqrt{7}$cm

(3) $12 \times \sqrt{\frac{5}{3}}$cm

(4) $\frac{12 \times 3}{\sqrt{7}}$ cm

62) ***Assertion :-*** In optical fibre, the diameter of the core is kept small. ***Reason :-*** This smaller diameter of the core ensures that the fibre should have incident angle more than the critical angle required for total internal reflection.

(1) Both assertion & reason are true & the reason is a correct explanation of the assertion.

(2) Both assertion & reason are true but reason is not a correct explanation of the assertion.

(3) Assertion is true but the reason is false.

(4) Both assertion & reason are false.

63) When light rays are incident on a prism at an angle of 45°, the minimum deviation is obtained. If refractive index of the material of prism is $\sqrt{2}$, then the angle of prism will be :

(1) 30°

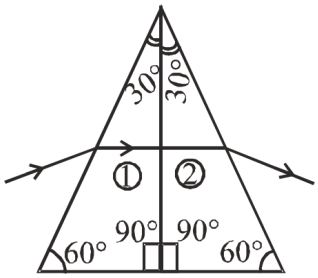
(2) 40°

(3) 50°

(4) 60°

64) Two identical prisms 1 and 2, each with angles of 30°, 60° and 90° are placed in contact as shown in figure. A ray of light passes through the combination in the position of minimum deviation and suffers a deviation of 30°. If the prism 2 is removed, then the angle of deviation for the same

ray is :



(1) equal to 15°

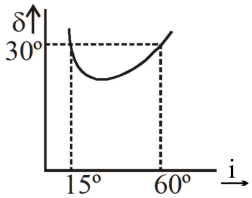
(2) smaller than 15°

(3) more than 15°

(4) equal to 30°

65) Figure shows graph of deviation (δ) versus angle of incidence (i) for a light ray striking a prism.

Angle of prism is :-



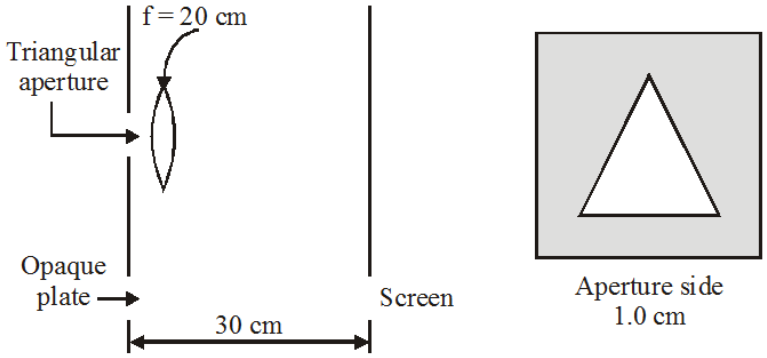
(1) 30°

(2) 45°

(3) 60°

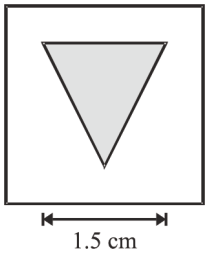
(4) 75°

66) A parallel beam of light falls axially on a thin converging lens of focal length 20 cm. The emergent light falls or a screen placed 30 cm beyond the lens. An opaque plate with a triangular aperture, side 1 cm, is in contact with the lens.

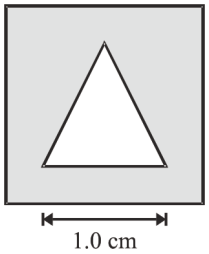


Which one of the following diagrams best shows to appearance of the patch of light seen on the screen?

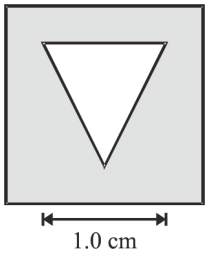
(1)



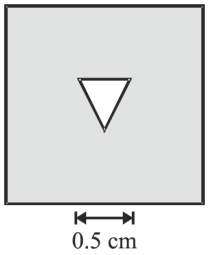
(2)



(3)



(4)



67) A beam of light converge at a point P. Now a lens is placed in the path of the convergent beam 12 cm from P. If the lens is a convex lens of focal length 20 cm the beam converge at a point (with respect to lens) :-

(1) –15cm

(2) +7.5cm

(3) –20cm

(4) +15cm

68) The radius of the convex surface of plano–convex lens is 20 cm and the refractive index of the material of the lens is 1.5. The focal length of the lens is

(1) 30 cm

(2) 50 cm

(3) 20 cm

(4) 40 cm

69) The focal length of a convex lens is 10 cm and its refractive index is 1.5. If the radius of curvature of one surface is 7.5 cm, the radius of curvature of the second surface will be :-

(1) –7.5 cm

(2) –15.0 cm

(3) –75 cm

(4) –5.0 cm

70) A convex lens of focal length 40 cm is in contact with a concave lens of focal length 25 cm. The power of combination is :-

(1) –1.5D

(2) –6.5 D

(3) +6.5 D

(4) +6.67 D

71) Lenses of powers 3D and –5D are combined to form a compound lens. An object is placed at a distance of 50 cm from this lens. Calculate the position of its image :-

(1) –10 cm

(2) +10 cm

(3) –25 cm

(4) + 25 cm.

72) A compound microscope consists of an objective lens of focal length 2 cm and an eye piece of focal length 6.25 cm separated by a distance of 15 cm. How far from the objective should one object be placed in order to obtain the final image at the least distance of distinct vision (25 cm).

(1) 2.5 cm

(2) 1.67 cm

(3) 2 cm

(4) 3.3 cm

73) The compound microscope is made from two lenses. Which statement is true concerning the operation of the compound microscope?

(1) Both lenses form real images

(2) Both lenses form virtual images

(3) The lens closest to the object forms a virtual image ; the other lens forms a real image

(4) The lens closest to the object forms a real image; the other lens forms a virtual image

74) The magnification produced by an astronomical telescope for normal adjustment is 10 and the length of the telescope is 1.1 m. The magnification when the image is formed at least distance of distinct vision (D = 25 cm) is :-

(1) 14

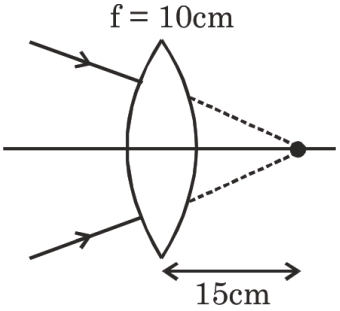
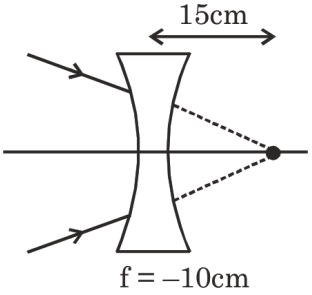
(2) 6

(3) 16

(4) 18

75) In the column-I, we have certain optical system and object. Match column-I with the size & nature of image formed from column-II :-

List–I	**List–II**

(P)		(1)	Magnified and virtual image.
(Q)		(2)	Diminished and virtual image
		(3)	Magnified and real image
		(4)	Diminished and real image.

(1) P-4; Q-1

(2) P-2; Q-1

(3) P-1; Q-2

(4) P-3; Q-1

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	**2**	**4**	**3**	**1**	**2**	**3**	**2**	**3**	**1**	**3**	**2**	**1**	**4**	**1**	**4**	**1**	**3**	**1**	**1**	**3**
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	**1**	**4**	**4**	**2**	**2**	**4**	**1**	**3**	**1**	**2**	**1**	**2**	**1**	**3**	**3**	**2**	**2**	**1**	**2**	**4**
Q.	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	**4**	**2**	**2**	**3**	**3**	**2**	**1**	**4**	**2**	**4**	**3**	**3**	**1**	**1**	**2**	**1**	**4**	**3**	**3**	**4**
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75					
A.	**4**	**1**	**4**	**1**	**2**	**4**	**2**	**4**	**2**	**1**	**3**	**1**	**4**	**1**	**1**					

SOLUTIONS

PHYSICS

1) w = f × S × cos 60º
= mgsin30° × S × cos 60º
$= 2 \times 9.8 \times \frac{1}{2} \times 2 \times \frac{1}{2}$
= 9.8 J

2)

(Mg sinθ + μ mg cosθ) $\sqrt{h^2 + \ell^2}$
mg $\left(\frac{h}{\sqrt{h^2 + \ell^2}} + \frac{\mu \ell}{\sqrt{h^2 + \ell^2}} \right) \sqrt{h^2 + \ell^2}$
$\frac{mg}{\sqrt{h^2 + \ell^2}}$ (h + μ□) $\sqrt{h^2 + \ell^2}$.

3) $F = \frac{K}{V} \Rightarrow FV = K \Rightarrow P = K$
$\frac{dW}{dt} = K$
$\int dW = k \int dt$
W = kt

4) Work = $\int Fdx = \int madx$
W = = m × Area under the curve a–x
$W = 5 \times \frac{1}{2} \times 20 \times 10^{-2} \times 8 \times 10^{-2}$
$W = 4 \times 10^{-2}$ J

5) F = 6t
⇒ ma = 6t
$\Rightarrow m\frac{dv}{dt} = 6t$
$\Rightarrow dv = \frac{6}{m}(t)dt$
$\Rightarrow \int_u^v dv = (6) \int_0^1 (t)\,dt$ (□ m = 1kg)
$\Rightarrow v = \frac{6}{2}\left(t^2\right)_0^1 = 3m/sec$
By work energy theorem

$$w = \frac{1}{2}m\left(v^2 - u^2\right) = \frac{1}{2}(1)\left[3^2 - 0^2\right] = 4.5J$$

6) Linear momentum of system of particle is '0'.
= either $\left(m_1\vec{v}_1 + m_2\vec{v}_2 + ... + m_n\vec{v}_n = 0\right)$ or $|\vec{v}_1| = |\vec{v}_2| = |\vec{v}_3| = |\vec{v}_n| = 0$
K.E. of system of particle is '0'.
then each particle must be at rest.
$|\vec{v}_1| = |\vec{v}_2| = |\vec{v}_3| ..... |\vec{v}_n| = 0$
So, option (C) is correct
If $(K.E.)_{system} = 0$ then $9\left(\vec{P}\right)_{system}$ must be '0'.

7) $x = (t - 3)^2$
velocity, $v = 2(t - 3)$
at $t = 0$, $v = -6$,
at $t = 6$, $v = 6$
Kinetic energy at $t = 0$ and $t = 6$ will be same,
so work done = ΔK.E. = 0.

8) $O = v_0^2 - 2ad \Rightarrow v_0 = \sqrt{2ad}$
$$v^2 = v_0^2 - 2a\frac{d}{4} = 2ad - \frac{ad}{2} = \frac{3ad}{2}$$
$$v = \frac{\sqrt{3}}{2}v_0$$
Alternative method
$$-Fd = O - \frac{1}{2}mv_0^2$$
$$-F\frac{d}{4} = \frac{1}{2}mv^2 - \frac{1}{2}mv_0^2$$
on solving $v = \frac{\sqrt{3}}{2}v_0$

9)

$W = \Delta K = K_f - K_i = K_f - 0$
area of F v/s t = $P_f - P_i$
$40 - 10 = mv_f - 0$
$30 = P_f$
$$K_f = \frac{P_f^2}{2m} = \frac{900}{2 \times 5} = 90\,J$$

10)

Spring force is a conservative force but does not follow inverse square law
so option (3) is false.

11) $U = -\int \vec{F}.d\vec{S}$

$U = -\int (4xydx + 2x^2dy)$

$U = -\int d(2x^2y)$

$U = -2x^2y + c$

12) $W = \frac{mg\ell}{2n^2}$

$n = \frac{5}{2}$

$W = \frac{2mg\ell}{25}$

13) Velocity at B when dropped from A, where AC = S

$v^2 = 0 + 2g(S - x)$ ...(i)

or $v^2 = 2g(S - x)$ ...(ii)

Potential energy at B = mgx

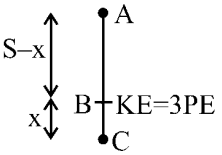
 Kinetic energy = 3 × potential energy

 $\frac{1}{2}m \times 2g(S - x) = 3 \times mgx$

or S – x = 3x or S = 4x or x = S/4

From eqn. (i), v2 = 2g(S – x) = $2g\left(S - \frac{S}{4}\right)$

$= \frac{2g \times 3S}{4} = \frac{3gS}{2}$



or $v = \sqrt{\frac{3gS}{2}}$

 $x = \frac{S}{4}$ and $v = \sqrt{\frac{3gS}{2}}$.

14) $\frac{1}{2}kx^2 = \frac{1}{2}mv^2$

$600 \times 9 \times 10^{-4} = 15 \times 10^{-3} \times v^2 \Rightarrow v = 6$ m/s

15) $\eta = \frac{p_0}{p_i} = \frac{mgh/\Delta t}{10kw} = \frac{9000 \times 10 \times 10}{5 \times 60} \times \frac{1}{10^4}$

= 0.3 ⇒ 30%

16) $x = t^2 + 2t - 1$

$v = 2t + 2$

$v_i = 2$ m/sec

$v_f = 4$ m/sec

$$P_{AV} = \frac{\Delta W}{\Delta t} = \frac{\Delta K.E.}{\Delta t}$$
$$= \frac{\frac{1}{2}(1)[4^2 - 2^2]}{1} = 6W$$

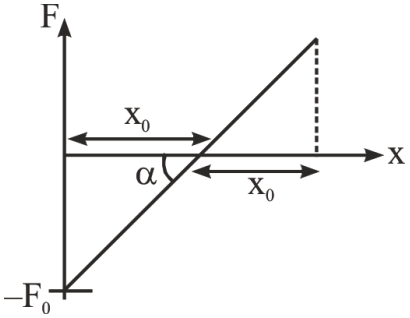
17)

$W_c = -\Delta U_{syc.}$
ΔU_{sys} does not depend on frame of reference.

18)

Work done = 0
$\Rightarrow$ Area = 0
$x = 2x_0$
Now $\frac{F_0}{x_0} = \tan\alpha$

$x^0 = \frac{F_0}{\tan\alpha}$ So $x = \frac{2F_0}{\tan\alpha}$ Hence option (1)



19) $\frac{p}{K} \propto \frac{1}{t} \Rightarrow \frac{\sqrt{2mK}}{K} \propto \frac{1}{t}$
$\therefore \sqrt{K} \propto t \quad \therefore K \propto t^2$
$\Rightarrow v^2 \propto t^2 \therefore v \propto t$
so, a = constant

20)

By WET (Ground to Gound)
$W_{gravity} + W_{Air} = \Delta KE$
$0 + W_{Air} = \frac{1}{2}m[v^2 - u^2]$
$W_{Air} = \frac{1}{2}(1)\ [(10)^2 - (16)^2]$
$= \frac{1}{2}(-6)\ (26)$
–78 J

21) $x_{cm} = \frac{m_1x_1 + m_2x_2 + m_3x_3}{m_1 + m_2 + m_3}$

$$x_{cm} = \frac{2 \times 0 + 4 \times \frac{1}{2} + 2 \times 1}{4 + 2 + 2} = \frac{1}{2}$$

$$y_{cm} = \frac{m_1 y_1 + m_2 y_2 + m_3 y_3}{m_1 + m_2 + m_3}$$

$$y_{cm} = \frac{2 \times 0 + 4 \times \frac{\sqrt{3}}{2} + 2 \times 0}{2 + 4 + 2} = \frac{\sqrt{3}}{4}$$

$(x_{cm}, y_{cm}) = \left(\frac{1}{2}, \frac{\sqrt{3}}{4}\right)$

22) $x_{cm_f} = x_{cm_i} + v_{cm} \times t$ and $a_{COM} = 0$

Here $v_{cm} = \frac{4 \times 2 - 4 \times 2}{4 + 4} = 0$

$x_{cm} = \frac{4 \times 0 + 4 \times 4.5}{4 + 4} = 2.25\text{m}$

$y_{cm} = 0$

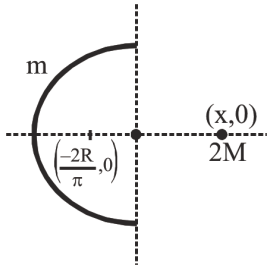
23)

$m_g v_g = m_b v_b$

$\frac{K_g}{K_b} = \frac{\frac{1}{2} m_g v_g^2}{\frac{1}{2} m_b v_b^2}$

$\because \frac{v_g}{v_b} = \frac{m_b}{m_g}$

$\therefore \frac{K_g}{K_b} = \frac{m_g}{m_b} \times \frac{m_b^2}{m_g^2} = \frac{m_b}{m_g}$



24)

$$x_{cm} = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}$$

$$0 = \frac{m\left(-\frac{2R}{\pi}\right) + 2M(x)}{3M}$$

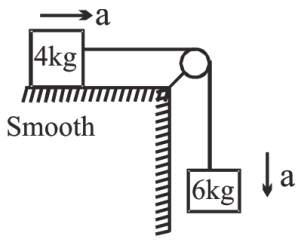
$$m\left(\frac{2R}{\pi}\right) = 2m \times x \Rightarrow \frac{R}{\pi} = x$$

$$x_{cm} = \frac{\int_0^L \left(\frac{kx^3}{L^2}\right) x\,dx}{\int_0^L \frac{kx^3}{L^2} dx} = \frac{4L}{5}$$

25)

26) $\overrightarrow{R_{cm}} = \frac{A_1\vec{r_1} - A_2\vec{r_2}}{A_1 - A_2}$

$= \frac{4m \times 0 - m \times \frac{a}{2}}{4m - m} = -\frac{a}{6}$

27) 

$\vec{a} = \frac{f_{net}}{M} = \frac{60}{4+6}$

$\vec{a} = 6 m/s^2$

$\vec{a}_{cm} = \frac{m_1\vec{a}_1 + m_2\vec{a}_2}{m_1 + m_2} = \frac{4 \times 6\hat{i} + 6\left(-6\hat{j}\right)}{4+6}$

$\vec{a}_{cm} = \left(2.4\hat{i} - 3.6\hat{j}\right) m/s^2$

28)

$a_{cm} = \frac{F_{ext}}{M_{total}} = \frac{6}{5}$

$V_{cm} = 0 + \left(\frac{6}{5}\right)5$

$V_{cm} = 6$ m/s

29) $\vec{P}_i = \vec{P}_f$

$0 = mv\hat{i} + mv\hat{j} + mv\hat{k} + m\vec{v}_4$

$\vec{v}_4 = -v(\hat{i} + \hat{j} + \hat{k})$

$|\vec{v}_4| = v\sqrt{3}$

$E_1 = \frac{1}{2}mv^2 = E_0; E_2 = \frac{1}{2}mv^2 = E_0;$

$E_3 = \frac{1}{2}mv^2 = E_0$

$E_4 = \frac{1}{2}mv_4^2 = \frac{1}{2}m \times (v\sqrt{3})^2 = 3E_0$

Total

$\Rightarrow E_1 + E_2 + E_3 + E_4 = E_0 + E_0 + E_0 + 3E_0 = 6E_0$

30)

$m_A = 90kg$, $m_B = 40$ kg, $m_C = 20$ kg

Let plank moves by x along left.

$m_A(\square - x) + m_B(-\square - x) + m_C(\square - x) = 0$

$(m_A - m_B)\square = (m_A + m_B + m_C)x$

$\Rightarrow (90 - 40) \times 12 = (90 + 40 + 20)x$

x = 4m

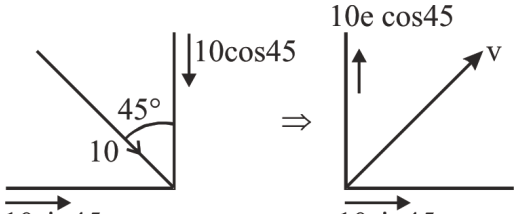
31) By conservation of linear momentum
(0.6) (4) + (0.4) (–2) = (0.6) (vA) + (0.4) (3)
$\Rightarrow v_A = \frac{2}{3}$ & $e = \frac{v_{sep}}{v_{app}} = \frac{3 - v_A}{4 + 2} = \frac{7}{18}$

32) Linear momentum depends on frame of reference as velocity depends on frame of reference.

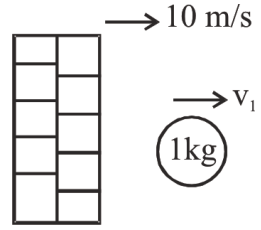
33) $mu = mv_1 + 2mv_1$
$\boxed{v_1 = u/3}$

$e = \frac{v_2 - v_1}{u_1 - u_2} = \frac{\frac{24}{3} - \frac{4}{3}}{u}$

$= \frac{u/3}{\mu}$ $\quad e = \frac{1}{3}$



34) $v = \sqrt{(10\sin 45)^2 + (10e\ \cos 45)^2}$
$v = 5\sqrt{3}$ m/s



35) $e = \frac{v_1 - 10}{10} = 1$
$v_1 = 20$ m/s

36) $\frac{1}{2} \times \left[\frac{4 \times 1}{4 + 1}\right] [6 - 1]^2 \left[1 - \frac{1}{4}\right]$
$\frac{1}{2} \times \frac{4}{5} \times 25 \times \frac{3}{4}$
$\frac{1}{2} \times 5 \times 3 = \frac{15}{2} = 7.5$ J

37) $m_1\overrightarrow{\Delta y_1} + m_2\overrightarrow{\Delta y_2} = 0$

$\vec{S}_{cm} = 0$

Let balloon descend by 'y'

⇒ M(–y) + m (L–y) = 0

⇒ –My + mL – my = 0

⇒ mL = my + My

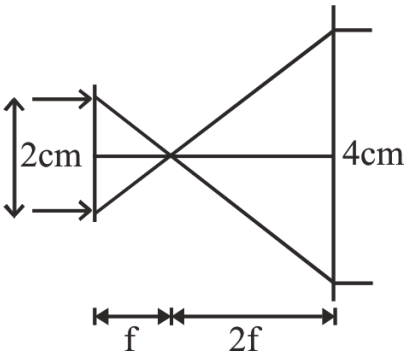
⇒ $y = \frac{mL}{(m + M)}$

38) In explosion of a bomb or inelastic collision between two bodies as force is internal, momentum is conserved while KE changes. hence, KE of a system can be changed without changing its momentum. Similarly, the reverse is also true, e.g., if a force acts perpendicular to motion work done will be zero and so KE will remain constant. However, the force will change the direction of motion and so the momentum.
Further, body may have energy (i.e., potential energy) without having momentum.

39) $\frac{\Delta K}{K} = \frac{2mgh}{8mgh} = \frac{1}{4}$

40) The initial velocity of CM is upward. The acceleration of the CM is 'g' downward.

41) Parallel and 2cm cross section radius. This is solved by sketching a ray diagram and using similar triangles. Ray emerge parallel to the optical axis and the emergent beam has twice the radius of incident beam



42) Magnifying power of astronomical telescope

$$m = -\frac{f_o}{f_e}\left(1 + \frac{f_e}{D}\right) = -\frac{200}{5}\left(1 + \frac{5}{25}\right) = -48.$$

43) $\frac{\text{Lateral shit}}{t} = \frac{\sin(i-r)}{\cos r}$

$= \frac{\sin i\cos r - \cos i\sin r}{\cos r}$

⇒ tan r = 1 – $\sqrt{\frac{2}{3}}$

44)

At crossing point it is C so
2f = 50
f = 25 cm

45)

Theory Based

46)

f = –10 cm $A_{obj.} = 9\ cm^2$
$A_{image} = m^2 A_{obj.}$

$$A_{image} = \left(\frac{f}{f-u}\right)^2 \times 9$$

$$\Rightarrow A_{image} = \left(\frac{-10}{-10+25}\right)^2 \times 9$$

$$A_{image} = \frac{4}{9} \times 9 = 4\ cm^2$$

47) Focal length of plane mirror = ∞

$$P_{plane\ mirror} = -\frac{1}{f} = \frac{1}{\infty} = 0$$

It is independent of medium.

48) $P_m = \dfrac{-100}{f(cm)} = +5D$

$$\frac{-100}{5} = f = -20\,cm$$

R of concave mirror = 2F = –40 cm

49) The refractive index of water with respect to air

$$\mu = \frac{C_{air}}{C_{water}} = \frac{v\lambda_{air}}{v\lambda_{water}},\ \mu = 1.33$$

$$\therefore \lambda \text{ in water} = \frac{\lambda_{air}}{\mu} \Rightarrow \lambda \text{ in water} = \frac{589}{1.33} nm$$

$\Rightarrow$ λ in water = 443 nm.

50) From graph it is clear that $\tan 30^o = \dfrac{\sin r}{\sin i}$

$$\Rightarrow \frac{1}{\sqrt{3}} = \frac{\sin r}{\sin i} = \frac{1}{\mu} \Rightarrow \mu = \sqrt{3}$$

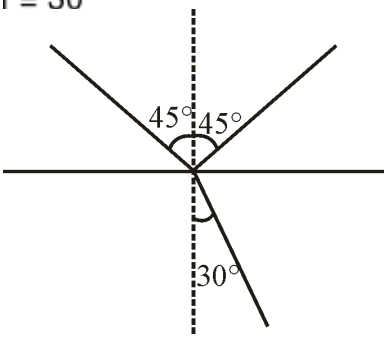
51) $_1\mu_2 = \frac{1}{\sin C} \Rightarrow \frac{\mu_2}{\mu_1} = \frac{\lambda_1}{\lambda_2} = \frac{1}{\sin C}$

$\Rightarrow \frac{6000}{4000} = \frac{1}{\sin C} \Rightarrow C = \sin^{-1}\left(\frac{2}{3}\right)$

52) $\mu = \frac{\sin i}{\sin r}$

$\sqrt{2} = \frac{\sin 45^\circ}{\sin r}$

$r = 30^\circ$



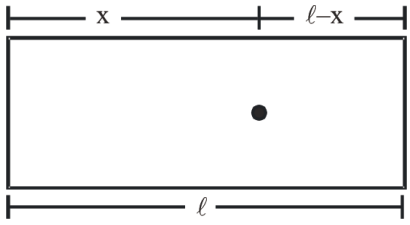
Angle = 180 – 45 – 30

= 105°

53) $\mu = \frac{c}{v} = \frac{3 \times 10^8}{2 \times 10^8} = 1.5$

$d_{ac} = 6$ cm

$x = d_{ac}\left(1 - \frac{1}{\mu}\right) = 6\left(1 - \frac{1}{1.5}\right) = 2cm$



54) $\frac{x}{\mu} = 5cm$ .....(i)

$\frac{\ell - x}{\mu} = 3cm$ .....(ii)

From (i) and (ii)

$\ell = (5+3)\mu = 12cm$

55) $\frac{H}{\mu} = (28 - \mu)$ (where H is height of water coloumb)

$\Rightarrow \frac{34}{4} + H = 28 \Rightarrow H = 16$ cm.

56)

$\mu = \frac{h_{act}}{h_{app}}$

$\frac{4}{3} = \frac{30}{h_{app}} \Rightarrow h_{app} = 22.5$ cm

Rain = 30 – 22.5 = 7.5 cm

57)

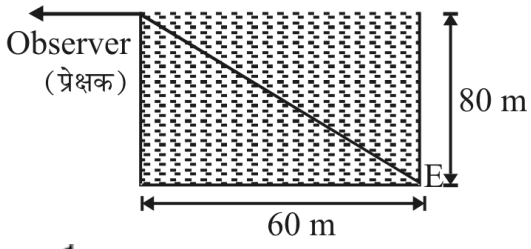
$\sin\theta_c = \frac{1}{\mu} \propto \lambda$

 as $\lambda\downarrow$, $\theta_c\downarrow$

λ_{min} for violet

$\therefore \theta_c$ min for violet

58)



$\sin\theta_C = \frac{1}{\mu}$

$\frac{60}{100} = \frac{1}{\mu}$

$\mu = 1.67$

59)

Snell's law gives $\mu_1 \sin 30° = \mu_2 \sin 60°$ but critical angle

$$\theta_C = \sin^{-1}\left(\frac{\mu_2}{\mu_1}\right) = \sin^{-1}\left(\frac{\sin 30^0}{\sin 60^0}\right) = \sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

60) T.I.R. can occur from A to B i.e. $\mu_A > \mu_B$

B to C i.e. $\mu_B > \mu_C$

$\mu_A > \mu_B > \mu_C$

$\frac{1}{\sin C_1} > \frac{1}{\sin C_2} > \frac{1}{\sin C_3}$

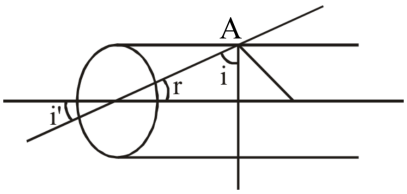
$\sin C_1 < \sin C_2 < \sin C_3$

$C_1 < C_2 < C_3$

61) Radius of circle, $r = \frac{h}{\sqrt{\mu^2 - 1}} = \frac{12}{\sqrt{\left(\frac{4}{3}\right)^2 - 1}}$

$= \frac{12 \times 3}{\sqrt{7}}$ cm

62) For smaller diameter incident angle at A(i) will be greater than critical angle to cause total internal reflection.



63)

$\sin 45^{o} = \mu \sin r_1$

$\frac{1}{\sqrt{2}} = \sqrt{2} \sin r_1$

$\frac{1}{2} = \sin r_1$

$r_1 = 30^{o}$

$A = 2r_1 = 60^{o}$

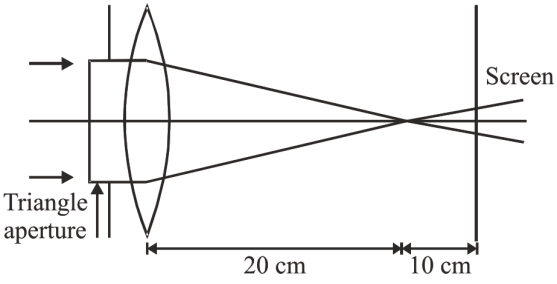
64)

$\delta = (\mu - 1) A$

As A is halved, Ao d is also halved.

65) $\delta = i_1 + i_2 - A$

$30° = 15 + 60 - A$

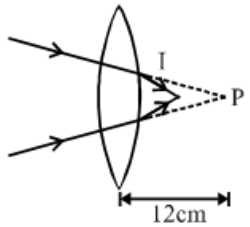


66) The image on the screen is real and inverted. The size of the image on the screen has aperture size given by

$$\text{Size} = 1.0 \left(\frac{10}{20}\right) = 0.5 \text{ cm}$$

67) $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

$\frac{1}{v} - \frac{1}{(+12)} = \frac{1}{20}$



$\boxed{v = +7.5 \text{ cm}}$

68)

$f = \frac{R}{\mu - 1} = \frac{20}{1.5 - 1} = 40$ cm

69) $\frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$

$\frac{1}{10} = (1.5 - 1)\left(\frac{1}{7.5} - \frac{1}{R_2}\right)$

$R_2 = -15$ cm

70) $\frac{1}{F_{eq}} = \frac{1}{40} - \frac{1}{25} = \frac{5 - 8}{200}$

$\Rightarrow F_{eq} = -\frac{200}{3}$

$P = \frac{100}{F_{eq}} = \frac{100}{-200} \times 3 = -1.5$ D

71) $P_{net} = P_1 + P_2$

$= 3 - 5 = -2$D

$f = \frac{-100}{2} = -50$ cm

$\frac{1}{v} - \frac{1}{-50} = \frac{-1}{50} \Rightarrow \quad v = -25$ cm

72) $f_0 = 2$cm $\quad f_e = 6.25$ cm $\quad L = 15$ cm

$v_e = -25$ cm

$\frac{1}{v_e} - \frac{1}{u_e} = \frac{1}{f_e} \Rightarrow -\frac{1}{25} - \frac{1}{u_e} = \frac{1}{6.25}$

$u_e = -5$ cm

Now,

$L = v_0 + |u_e|$

$15 = v_0 + 5 \quad \Rightarrow \quad v_0 = 10$ cm

Now, for objective

$\frac{1}{v_0} - \frac{1}{u_0} = \frac{1}{f_0} \Rightarrow \quad \frac{1}{10} - \frac{1}{u_0} = \frac{1}{2}$

$\boxed{u_0 = -2.5 \text{ cm}}$

73) Objective lens forms real image, eyepiece forms a virtual image.

74) $M = \frac{f_0}{f_e} = 10$

$f_0 + f_e = 1.1$

$f_0 = 100$ cm , $f_e = 10$ cm

$$M = \frac{f_0}{f_e}\left(1 + \frac{f_e}{D}\right) = \frac{100}{10}\left(1 + \frac{10}{25}\right) = 14$$

75)

(P) $\frac{1}{f} = \frac{1}{v} - \frac{1}{u} \Rightarrow \frac{1}{(+10)} = \frac{1}{v} - \frac{1}{(+15)}$

$\frac{1}{v} = \frac{1}{10} + \frac{1}{15} \Rightarrow \frac{1}{v} = \frac{1}{6}$

v = + 6 cm (real image)

Now magnification

$m = \frac{v}{u} = \frac{(+6)}{(+15)} = +\frac{2}{5} = 0.4$ (Diminished image)

(Q) $\frac{1}{f} = \frac{1}{v} - \frac{1}{u} \Rightarrow \frac{1}{(-10)} = \frac{1}{v} - \frac{1}{(+15)}$

$\frac{1}{v} = \frac{1}{15} - \frac{1}{10} \Rightarrow \frac{1}{v} = \frac{-1}{30}$

v = – 30 cm (Virtual image)

Now magnification

$m = \frac{v}{u} = \frac{(-30)}{(+15)} = -2$ (Magnified image)